ISyE 6202 & ISyE 6335 Supply Chain Facilities

Casework 3: StackBay Warehousing Storage Design — Phase 1

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- *The casework is meant to be realized in self-organized teams of up to 4 students, yet may be realized solo, with the number of students in the team having no influence on the grading.*
- *The casework has a weight of 30 points.*
- *The casework must be completed and the report delivered by November 16th.*
- *The casework is shaped around a set of mandatory tasks having a significant grading weight, purposefully not revealed to the class.*
- *For each task, you must rigorously describe your methodology, present your results, analyze them, and discuss them relative to the current task and, when pertinent, relative to previous tasks.*
- *For computational algorithms and simulations, exploitation of Python and linked tools is preferred. Using AI prompting as support enabler is allowed. Using spreadsheets such as Microsoft Excel/VBA is tolerated. All developed codes must be provided. For maps and visualizations, exploiting Python libraries or Tableau (or similar analytics software) is recommended.*
- *The answers must be limited to a maximum of 3 pages per task. Evaluation results are not affected by task answer length, rather by the excellence of answers.*
- *TA Mina Kim will be main one responsible for supporting the teams in this casework. She will interact directly with Ph.D. candidate Sahrish Shaikh and Prof. Montreuil. As usual the other TAs and Prof Montreuil will be available to answer questions during their office hours.*

Context

StackBay is a fictional third-party warehousing logistics service provider. From a single dry-goods warehouse in the US Southeast, it stores and ships full pallets of food, packaging, beverages, and supplies for nine restaurant chains, its clients. The clients own the stock. StackBay receives it from their suppliers, keeps it in pallet racking, and delivers it to the clients' restaurants on its own trucks.

StackBay is paid per pallet handled and per pallet-day stored. Its margin therefore depends on two things: how many operating resource hours it spends per pallet, and how well it uses the building. The warehouse has grown with its clients. Its storage policy was never conscientiously designed. Put-away follows practices that drivers have built up over the years, not a written rule, and nobody has measured what they cost. Management suspects that a large share of forklift time is spent driving pallets to and from the far end of the building, and that tall rack openings are filled with short pallets while tall pallets require looking for room.

Management now wants an outside view. It wants to know how well the warehouse really performs today, which storage policy it should adopt, whether it is worth moving the pallets already in stock or changing the height of the rack beams to adopt it, and how many forklifts it needs and how many forklift drivers should be active on each shift. As this is important to StackBay, it has decided to work with a number of teams, enabling it to get and contrast alternative proposals. You are essentially competing with other teams, so leverage your creativity, rigor, knowledge, and pragmatism.

StackBay is explicit about one thing: it wants a systematic understanding of its operation before any change is proposed. A proposal that scores well but cannot explain why will not be adopted. The tasks below are ordered accordingly: understand the business, diagnose the warehouse, measure it, and only then design, test and recommend.

Clients and Stores

StackBay serves nine clients with 936 restaurants across seven states. Each product (SKU) belongs to exactly one client, and only that client's restaurants order it. The clients differ greatly in size, as shown below for the historical year.

Client	Stores	SKUs	Pallets shipped per week
Birch Restaurants	76	257	563
Alder Restaurants	352	229	542
Ironwood Restaurants	166	217	281
Cedar Restaurants	92	116	253
Dogwood Restaurants	63	138	135
Elm Restaurants	93	109	114
Fir Restaurants	86	63	25
Ginkgo Restaurants	5	34	11
Hazel Restaurants	3	26	11

Joined files will provide details on the following facts. Restaurant locations are given at the level of the nearest city. Small and medium restaurants are delivered once a week, the larger ones twice a week. Deliveries to specific restaurants are set at fixed day-s within the week, using a set of routes repeated every week.

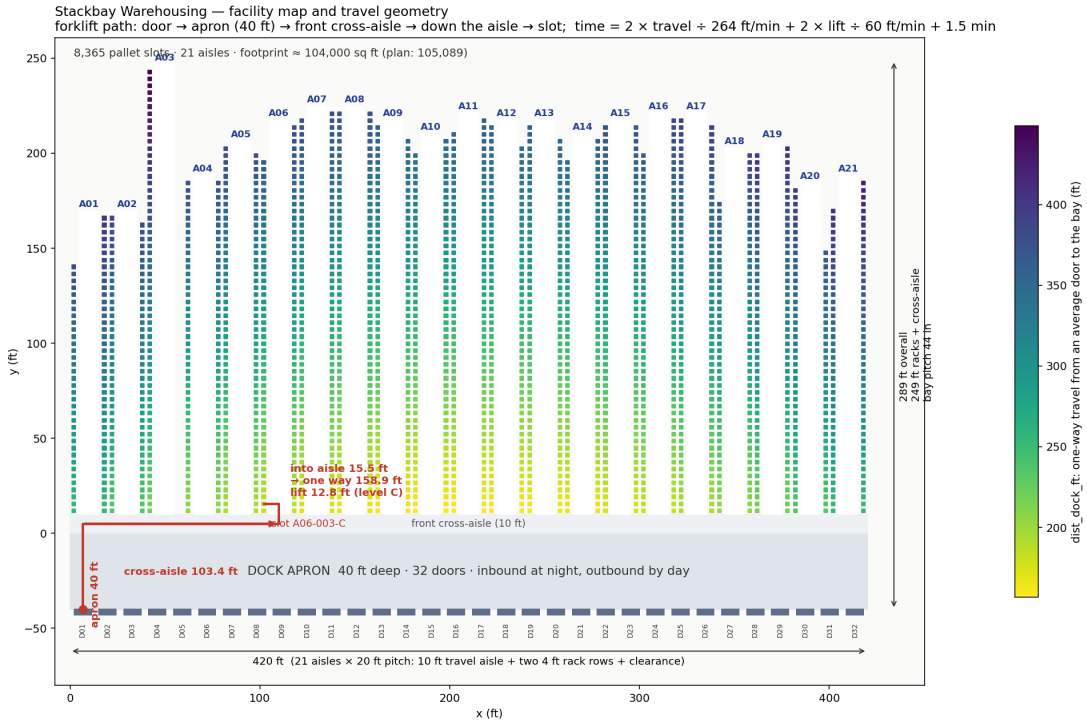


Figure 1. Facility map: aisles A01 to A21, slots colored by distance from the dock, the 32 dock doors, and one forklift path.

The Warehouse

Schematized in Figure 1, the warehouse is a single block of about 105,000 square feet: 21 rack aisles running back from one dock of 32 doors on the front wall. The racks hold 8,365 pallet slots on up to seven levels. Each slot takes exactly one pallet. Slot openings vary between 24 and 130 inches in height, so not every pallet fits every slot. The warehouse operates at ambient temperature as StackBay does not deal with cold chain products.

A forklift leaving a dock door crosses a 40-foot dock apron, drives along the front cross-aisle to the aisle it needs, then down the aisle to the bay. There is no cross-aisle at the back of the building: every trip into an aisle comes back out the same way. Each aisle and the front cross-aisle have one traffic lane in each direction.

Products and Pallets

The 1,189 SKUs are dry grocery, paper and disposables, beverages, cleaning products, and smallwares. Each is received and shipped as a full pallet of a fixed number of cases, its Ti × Hi (cases per layer × layers). Loaded pallets stand between 26 and 88 inches tall and weigh up to 2,400 pounds. Demand is highly concentrated: a few dozen SKUs make up half of all pallets shipped, while most SKUs ship only a handful of pallets a year.

Daily Operating Cycle

StackBay runs two shifts, in a fixed order every day:

- **Day shift, about 04:00 to 16:00, Sunday to Friday.** Forklift operators pick the day's pallets from the racks, stage them in a lane at their truck's door, and load them. Trucks leave between about 05:30 and 15:30, each on its own route, with up to 26 pallets. There is no shipping on Saturday.
- **Night shift, from about 16:30 until midnight, Monday to Friday.** Supplier trucks arrive, are unloaded into the staging lane at their door, and every pallet is put away into a slot before midnight. There is no receiving at the weekend.

A pallet received tonight can therefore ship tomorrow at the earliest. StackBay currently always ships the oldest pallet of a SKU first (first in, first out). A slot that is emptied stays empty until a later receipt fills it; pallets are never moved from slot to slot, and there is no separation between pick and reserve locations.

Forklift operators are typically scheduled to work 36 hours a week, in a dynamic assignment depending on loading and availability.

Replenishment

StackBay's buyers reorder each SKU from its supplier on the client's behalf, so that stock covers the expected demand until the next delivery with a safety margin. Each supplier delivers on fixed weekdays, from one to five a week depending on its volume, and one purchase order covers all of that supplier's SKUs that need stock replenishment. Supplier lead times range from 4 to 41 days, and deliveries are not always on time. Replenishment is out of scope for this casework: what arrives, and when, is given and is the same whatever storage design you propose.

Forklift Operations

Every pallet is currently handled four times by a forklift carrying one pallet per trip: picked from its slot to the staging lane at its truck's door, loaded from the lane into the trailer, and on the way in, unloaded from the supplier's trailer to the lane, then put away from the lane to its slot. This is driven by StackBay's prioritization of truck loading and unloading start-to-finish times, thus minimizing their time spent at a dock. Only one forklift works a truck at a time, loading or unloading. At most two forklifts work in an aisle at once, one in each direction. The day's picks are released at the start of the day shift and worked truck by truck in order of departure; a truck leaves at its scheduled time if its load is complete, otherwise as soon as the last pallet is on.

Storage Today

There is no written storage policy, but put-away is not random either: drivers follow established practices when they choose among the free slots that are tall enough for a pallet. What those practices are, what they take into account, what they ignore, and what they cost StackBay, is for you to establish from the data.

Modeling and Calculation Rules

These rules define calculations that must remain consistent across experiments.

1. **Travel distance.** The one-way forklift distance between a dock door and a slot equals 40 ft of dock apron, plus the distance along the front cross-aisle from the door to the aisle, plus the depth into the aisle to the bay. It is given for every historical pallet as travel_ft. dist_dock_ft in slots.csv is the same path averaged over the 32 doors.
2. **Lift.** The lift of a move equals the slot's height above the floor, given as lift_ft (elev_ft in slots.csv).
3. **Move time.** A slot move takes, in minutes, 2 × travel_ft ÷ 264 + 2 × lift_ft ÷ 60 + 1.5: the round trip at 3 mph, the lift up and down at 60 ft per minute, and one and a half minutes of handling for positioning, picking up or setting down, and scanning. A dock move between staging lane and trailer takes about 1.8 minutes.
4. **Labour hours.** Operating forklift labour hours per week equal the sum of the move times of all put-aways and picks, divided by 60, per week.
5. **Slot fit.** A pallet fits a slot when its height plus 4 inches of clearance does not exceed the slot's opening height (max_pallet_in = opening height − 4 in).
6. **Flows are fixed.** Receipts and shipments, their dates, times, trucks and doors are the same under every storage design. A design changes only where pallets are stored, hence travel, lift, fit, and labour. Stock levels and time in stock do not depend on the design.

The Data Package

Every team receives the same data package. All data are synthetic: the package is an engineered case inspired by a real kind of operation, not a description of any real company, and it holds no real clients, suppliers or personal data. The historical year runs from July 7th, 2025 to July 5th, 2026.

File	Contents	Released
slots.csv	Every pallet slot: aisle, side, bay, level, coordinates, aisle position, elevation, opening height, tallest pallet accepted, cube, average distance to the dock.	At start
docks.csv	The 32 dock doors and their positions.	At start
skus.csv	Every SKU: client, category, Ti, Hi, cases per pallet, case cube, pallet height and weight, supplier.	At start
clients.csv, client_sites.csv	The nine clients; every restaurant with its client and nearest city.	At start
suppliers.csv	Suppliers with their number of SKUs and lead time.	At start
opening_stock.csv	The pallets in the racks on 7 July 2025, with SKU, slot and receipt date.	At start
inbound.csv	Every pallet received: date, arrival and put-away time, purchase order, supplier, door, SKU, slot, travel and lift.	At start

File	Contents	Released
outbound.csv	Every pallet shipped: date, pick and departure time, truck, route, door, stop, restaurant, client, SKU, slot picked from, travel and lift, receipt date, days in stock.	At start
trucks.csv	Every outbound truck: date, departure time, door, route, stops, pallets.	At start
current_stock.csv	The pallets in the racks on 5 July 2026, with SKU, slot and receipt date.	At start
Maps and overview	facility_map.png, floor_plan.png, store_map.png, and overview.html with summary charts.	At start

Fields that you are expected to derive, such as velocity classes, stock by SKU over time, peak stock, forklift workload and today's storage behavior, are not supplied as answers.

The StackBay Digital Twin

The StackBay Digital Twin replays any day of the historical year in three dimensions, hour by hour: trucks arriving and leaving at their doors, forklifts picking, loading, unloading and putting away, pallets waiting in the staging lanes, and the racks filling and emptying. You can choose the date and the time of day, slow the clock down to real time, color the pallets by client, by days in stock or by height fit, and change the number of forklifts on shift. With fewer forklifts, pallets queue, trucks wait for their load and the night shift runs past midnight. The twin is a sandbox for understanding the operation; it does not evaluate storage designs.

Access. The twin is a web page: open it at Stackbay Twin (no code needed). It is for exploring the operation and is not provided as a file.

The casework has nine mandatory tasks in three phases. This document describes Phase 1, Tasks 1 to 3; each task builds on the ones before it. Once your team has submitted its Phase 1 deliverables, you receive the description of Phases 2 and 3 (Tasks 4 to 9) and access to the StackBay Storage Design Evaluator.

PHASE 1

Task 1. Characterize the business

Describe what flows through the warehouse, from the released historical year, before considering the warehouse itself.

a. Characterize each client: stores, SKUs, pallets shipped per week, and where its restaurants are. Identify the clients that drive the operation and those that hardly register.

b. Rank the SKUs by pallets shipped and draw the cumulative curve. Define your own velocity classes with explicit cut-offs, justify them, and keep them for the later tasks. Characterize pallet heights and weights, notably for the fast movers.

c. Characterize demand over time: pallets per week across the year, per day of the week, and per hour of the day. Identify trend, seasonality and the peak days and hours, and quantify week-to-week variability.

d. Characterize the restaurants, routes and trucks: pallets per restaurant per week, delivery days, routes, trucks per day, pallets and stops per truck. Explain why some trucks leave full and others nearly empty.

e. Characterize supply: purchase orders per week, SKUs and pallets per order, the suppliers that matter, lead times, and the receiving nights. State what this implies for the night shift.

f. Reconstruct the number of pallets in the building every day of the year, in total and for each SKU, from the opening stock, receipts and shipments. Report the average and peak stock, in pallets, as a share of the slots, and in days of shipments. Report each SKU's average and peak stock.

Deliverable. A business profile with visuals and Tables, ending with the three facts about StackBay's flows that a warehouse designer must design around.

Task 2. Diagnose current operation

Describe how the warehouse currently works and whether that makes sense. Do not propose changes yet.

a. Describe the layout from slots.csv: aisles, bays, levels, slots per level, the distribution of opening heights and where the tall openings are. Verify the distances against the facility map.

b. Establish today's storage behavior from where pallets were actually put. Test whether storage location relates to SKU velocity, to client, to pallet height, or to none of these. Quantify the height left unused above stored pallets. State today's put-away rule in one paragraph, as a driver would describe it, and the evidence for each element of it.

c. Verify first-in first-out from the data. Characterize how long pallets stay, overall, by velocity class and by client, and identify the pallets that stayed longest.

d. Estimate from the data the forklift hours needed per day shift and per night shift, using the modeling rules. In the Digital Twin, vary the number of forklifts on shift on the same day and identify when trucks start waiting, when the night shift runs past midnight, and why adding forklifts stops helping beyond some point.

Deliverable. An operations diagnostic, with your statement of today's put-away rule, ending with the three aspects of current operations you consider most questionable and the evidence for each.

Task 3. Establish the performance baseline

Turn the diagnosis into measures that can be compared, now and later against your own designs.

a. Define the performance measures you will use, covering at least travel, labor, space, fit, and inventory, from the perspectives of StackBay, its clients and its forklift drivers. For each, state exactly how it is computed from the files, and whether and how a storage design can affect it. Identify the measures a storage design cannot affect, and why.

b. Compute every measure for the historical year as it was operated.

c. Break down forklift travel and labor by velocity class, client, rack level, and aisle. Quantify the share of all travel caused by the fastest 5% of SKUs.

d. Estimate how much travel and labor would have been saved had the fastest SKUs been stored in the slots closest to the dock. State every assumption.

e. Judge whether StackBay's operation is good or poor, separating what is inherent in the business and the building from what the warehouse chooses to do.

Deliverable. A baseline scorecard: the measure definitions, the baseline values, the travel breakdown, the preliminary estimate of the opportunity, and your judgement.

Minimum Reporting Standard

Report enough information for another team who has access to it being able to reproduce each result. For evaluator experiments, state the design files, the futures used and the run conditions. For searches or comparisons, state the alternatives, constraints and controlled inputs. Report every design on the same measures.

Phase 1 Submission

- The deliverables of Tasks 1 to 3: methods, results, discussion, assumptions and limitations.
- The velocity class and stock file from Task 1: one row per SKU with its pallets shipped, your velocity class, and its average and peak stock.
- All source code and models, with a README that reproduces your results.
- For teams, the contribution matrix, and the ethical statement relative to the usage of AI.

Access to Phase 2. When your team's Phase 1 submission has been received, you get the Phase 2 document, the link to the Storage Design Evaluator and your team's access code. Every evaluator run is recorded with your team, the time and the files uploaded.

We hope this casework proves to be a stimulating and worthwhile learning experience,

Benoit Montreuil and Sahrish Shaikh